

Samarth Narahari Mandagere

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EDUCATION

University of Southern California

Master of Science in Computer Science

- Relevant Coursework: Analysis of Algorithms, Software Engineering, Database Systems, Web Technologies

Los Angeles, CA

Aug 2025 – May 2027

BNM Institute of Technology

Bachelor of Engineering in Computer Science — GPA: 9.51 / 10

- Built award-winning AI and full-stack projects, conducted research focused on computer vision, NLP, and real-time systems

Bengaluru, India

Nov 2021 – Jun 2025

EXPERIENCE

Founding Software Engineering Intern

Vylar (LostStories App)

Sep 2024 – Feb 2025

Bengaluru, India

- Developed full-stack creator publishing workflows using **React, Node.js, Express, MongoDB/Mongoose, and Python/Flask** for audio uploads, transcription, metadata processing, and automated tag extraction
- Deployed separate frontend and backend **Microservices** on **AWS EC2 with Amazon Linux**, integrating React/Node.js application services with Python/Flask transcription APIs for production access
- Engineered content recommendation pipelines using **Natural Language Toolkit(NLTK) named-entity extraction** to identify people, places, dates, and topics from transcripts and recommend related stories based on shared entities
- Designed multilingual audio transcription pipelines using **OpenAI Whisper, SpeechRecognition, and pydub**, including silence-based audio chunking to reduce transcription latency from **3 hours to 12 minutes**

Software Engineering Intern – Computer Vision

Samsung PRISM

Jun 2023 – Dec 2023

Bengaluru, India

- Built an automated visual regression pipeline using **Python, OpenCV, and NumPy** to compare device screenshots against captured images for UI consistency verification
- Implemented **Structural Similarity Index Measure** with **scikit-image, template matching, image alignment, thresholding, and contour-based anomaly detection** to localize and highlight visual UI defects
- Optimized **batch image-processing** pipelines, reducing UI verification runtime from **15–20 minutes to under 1 minute**

PROJECTS

RoleReady – AI-Powered Onboarding Platform | Python, FastAPI, LangChain, Gemini, Supabase/PGVector, GCP

- Built a multi-source **RAG** platform ingesting organizational knowledge from **GitHub, Jira, Confluence, and stored context** using **Python/LangChain**, with **Supabase/PGVector** for semantic retrieval
- Engineered organizational context retrieval and structured onboarding content generation services using **FastAPI, PGVector semantic search, and Gemini**
- Developed end-to-end mindmap generation using **Gemini, Mermaid, React, and Supabase**, transforming RAG-retrieved context into structured JSON and interactive visual learning content
- Built an AI-driven video generation pipeline using **Gemini, MoviePy/FastForward-mpeg, text-to-speech, and Supabase Storage**, reducing generation time from **2 hours to 15 minutes** for 7-minute educational videos

PromptQuest – Gamified AI Prompting Platform | React, TypeScript, Flask, Gemini, Docker, GCP

- Architected and deployed a full-stack platform utilizing a **Vite/React/TypeScript frontend on Vercel** and a containerized **Flask/Gemini API** backend using **Docker** and **Gunicorn** on **Google Cloud Run**
- Engineered a real-time prompt evaluation and scoring engine for gamified AI challenges using **Flask REST APIs** and **Google Gemini** for accurate scores
- Redesigned deployment from Vercel serverless to split infrastructure using **Docker, Gunicorn, Google Cloud Run, and Vercel**, resolving Python runtime and routing failures
- Won **1st Place** at the Vercel Software Awards during LA Tech Week

Indian Sign Language Interpreter | Python, TensorFlow, MediaPipe, OpenCV, Flask, MongoDB

- Built a real-time Indian Sign Language recognition platform using **TensorFlow/Keras, MediaPipe, OpenCV, and NumPy**, extracting 3D hand landmarks for gesture and word classification
- Engineered a landmark-based inference pipeline using **MediaPipe Hands** and **scikit-learn** preprocessing, transforming multi-minute image recognition into real-time webcam-based prediction
- Integrated **NLTK** tokenization, edit-distance matching, and WordNet-based post-processing to convert recognized gesture sequences into context-aware words
- Developed a **Flask and MongoDB** web application with user authentication and progress tracking for interactive sign-language learning and recognition

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C/C++

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, Flask, LangChain, TensorFlow, OpenCV, MediaPipe, NumPy, scikit-learn

Databases & Cloud: PostgreSQL, MongoDB, Supabase/PGVector, Docker, AWS EC2, GCP, Google Cloud Run, Linux

Tools & Concepts: Git/GitHub, REST APIs, RAG, Vector Search, NLP, Machine Learning, Computer Vision